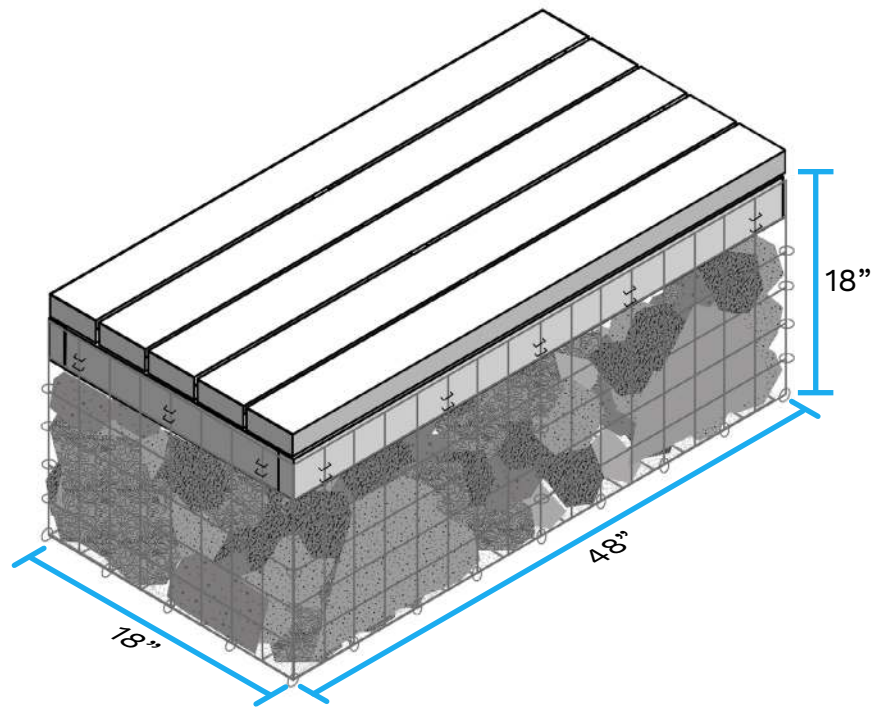




4' GABION BENCH

ASSEMBLY INSTRUCTIONS

BEFORE YOU BEGIN

1. This document includes recommendations for do-it-yourself (DIY) construction of a 4' gabion bench. As a DIY project, the Owner, or the Owner's Contractor, is required to procure, fabricate, assemble, install, and maintain all project materials. These are recommendations only and any material, assembly, and construction must be tailored to specific project requirements.
2. These recommendations are based on common best practices associated with standard DIY home-improvement and/or landscape construction. These best practices may include working knowledge and experience with manual and power tools, hardware, fasteners, lumber, concrete footings, and associated techniques of assembly and construction. Prior working knowledge of these best practices is required so as to avoid harm or personal injury to self or others, or to property.
3. This document includes recommendations for the sequence and assembly of gabion bench components. Locate your gabion benches in your plan you created in the 'Dream' workbook to find the proposed location and dimension of all gabion benches. This plan must be tailored to the specific conditions of the project site, including property lines, existing features- such as walls, trees, etc- and any other existing site conditions that may limit or otherwise influence the installation. Use your cost estimation spreadsheet to determine the quantity needed for all your gabion benches.
4. Before using any power equipment (such as drills, saws, augers, or other), review all manufacturer safety requirements. While handling or operating any power equipment, follow all manufacturer safety requirements. Do not allow others who have not reviewed and/or follow manufacturer safety requirements to handle or utilize power equipment.
5. The owner is solely responsible for any work required to secure, and comply with, all necessary building and safety codes, permits, and regulatory approvals.
6. These recommendations assume that the project site has been sufficiently prepared for construction. This includes, but is not limited to: identification of utilities that may interfere or otherwise impact installation; the removal of trash, hazards, unwanted structures; or other conditions that may inhibit safe construction.
7. Before construction, review the project site conditions to identify any potential limitations to assembly and installation. This includes determining advance notification by the Owner of utility companies so as to determine the location of all above and below ground utilities.
8. All persons associated with building and construction should sign a volunteer waiver.

LEGAL NOTICE:

The recommendations included herein are based on standard practices of fabrication and construction of garden furnishings. In no event shall Thomas Jefferson University (TJU), TJU's Lab For Social Innovation, Park-in-a-Truck, and/or its faculty, staff or affiliates be liable for any damages or injury to person or property damages arising out of the use of these recommendations. Thomas Jefferson University (TJU), TJU's Lab For Social Innovation, Park-in-a-Truck, or its faculty, staff and/or affiliates make no warranties, expressed or implied, and hereby disclaims and negates all other warranties, including without limitation, implied warranties or conditions of merchantability, fitness for a particular purpose, or non-infringement of intellectual property or other violation of rights. Further, Thomas Jefferson University (TJU), TJU's Lab For Social Innovation, Park-in-a-Truck, and/or its faculty, staff or affiliates do not warrant or make any representations concerning the accuracy, likely results, or reliability of the use of these recommendations. The use of these recommendations constitutes your acknowledgment of the aforementioned.

4' GABION BENCH

ASSEMBLY INSTRUCTIONS

COMMON MATERIAL SELECTIONS \

GABION BASKET

METAL 2X2 WIRE MESH, FABRICATED WITH 8 GAUGE (3/16" DIA. WIRE) WELDED-WIRE, GALVANIZED-WITH 2" OPENINGS.

FABRICATION SUGGESTIONS:

DARBY AND SONS: WWW.DARBYWIREMESH.COM

GABION1.COM: [HTTPS://GABION1.COM/PRODUCT/18-HIGH-X-18-THICK-1/](https://GABION1.COM/PRODUCT/18-HIGH-X-18-THICK-1/)

NOTE: WHEN ORDERING CUSTOM CUT PIECES FROM A METAL SUPPLIER LIKE DARBY AND SONS, ENSURE TO STATE THE PIECES SHOULD BE CUT TO SIZE, CLOSED PIECES, GALVANIZED AFTER

SIZES: SEE STEP 01

BRACING MATERIAL

OPT 1: ROLL OF 14 OR 16 GAUGE SMOOTH, GALVANIZED WIRE

OPT 2: EXCESS PIECES OF 2X2 WIRE METAL MESH (YOU WILL HAVE EXTRA PIECES OF WIRE MESH TO CUT IN HALF IF BUILDING MORE THAN ONE GABION BASKET AT A TIME)

GABION FILL MATERIAL

NOTE: STEPS 1-7 CAN BE PREFORMED OFF SITE BEFORE THEY ARE FILLED IN (STEP 9)

ALL FILL MATERIAL SHOULD BE ANGULAR IN SHAPE. MATERIALS SHOULD BE HARD AND INERT SO THAT IT WILL NOT BREAK-DOWN OVER TIME. BOTH THE FILL MATERIAL AND THE GABION PANELS WORK TOGETHER TO PROVIDE STRUCTURAL INTEGRITY. CUT LOGS OR ANY ORGANIC MATERIAL THAT CAN BREAK DOWN OVER TIME SHOULD NOT BE USED FOR THIS REASON.

MATERIALS FREQUENTLY USED FOR GABION FILL INCLUDE QUARRIED ROCK, RECYCLED PAVEMENT MATERIALS (CONCRETE, ASPHALT, OTHER) OR OTHER RECYCLED MATERIALS SUCH AS BRICK OR CONCRETE MASONRY UNITS. THE SELECTION OF APPROPRIATE FILL MATERIALS IS UP TO YOU.

FILL MATERIAL SHOULD BE A MIX OF SIZES, RANGING FROM 12 INCHES MAXIMUM TO 2 INCHES MINIMUM. ALL FILL MATERIAL PIECES USED ON THE EXTERIOR FACE WILL NEED TO BE LARGER THAN 3 INCHES.. SMALLER FILL PIECES WILL BE USED TO FILL IN ANY VOIDS, AS NECESSARY.

<https://www.rockler.com/learn/choosing-the-best-finish-for-outdoor-projects>
<https://www.realmilkpaint.com/blog/tips/best-finish-outdoor-wood/>
<https://vermontnaturalcoatings.com/product-category/exterior/>

PARK IN A TRUCK The Landscape Architecture Program + The Lab for Social and Urban Innovation Thomas Jefferson University

WORKING VERSION 1.0

4' GABION BENCH \

TOOLS:

10" OR 12" MITER SAW (or appropriate hand saw)

IMPACT DRIVER for self drilling exterior wood screws

DRILL with 1/8" drill bit

SANDPAPER 120 GRIT

CARPENTERS SQUARE

SAFETY GOOGLES + GLOVES

HOG RING PLIERS

BOLT CUTTERS

HAMMER

4lb SLEDGEHAMMER

GRINDER

MATERIALS:

(36) 2.5 SELF DRIVING EXTERIOR WOOD SCREWS

(50) 3/4" HOG RINGS

(32) CABLE STAPLES

COMMON LUMBER SELECTIONS \

WESTERN RED CEDAR:

PRO: NATURALLY WEATHER/ROT RESISTANT LUMBER WITH AESTHETICALLY PLEASING RED TONES AND PLEASANT AROMAS, THAT IS EASY TO WORK WITH. NO PRESERVATION TREATMENTS ARE NECESSARY AS THE SURFACES WILL GRADUALLY DEVELOP A LOVELY GRAY PATINA. IT IS EASILY RECYCLABLE AT THE END OF ITS PRIMARY USE.

CON: COST, BE SURE TO SPECIFY 'KNOTTY' (STK) GRADE WHICH IS MUCH MORE AFFORDABLE THAN 'CLEAR' GRADE.

DOUGLAS FIR:

PRO: MODERATELY WEATHER/ROT RESISTANT AND STRONG LUMBER. IT IS ALSO EASY TO WORK WITH AT A LOWER COST THAN CEDAR. IT IS EASILY RECYCLABLE AT THE END OF ITS PRIMARY USE.

CON: IT WILL NOT STAND THE TEST OF TIME AS WELL AS CEDAR, BUT A GREAT CHOICE ESPECIALLY IF YOU PLAN TO PAINT IT.

PRESSURE TREATED:

PRO: WEATHER/ROT RESISTANT LUMBER AT A LOW COST.

CON: LUMBER IS OFTEN OF VARIABLE POOR QUALITY AND SATURATED WITH CHEMICAL TREATMENT MAKING IT DIFFICULT TO WORK WITH LEADING TO INCONSISTENT RESULTS. NEEDS 1-4 MONTHS OF AIR DRYING BEFORE PAINTING. NOT EASILY RECYCLABLE AND OFTEN ENDS UP IN LANDFILLS AT THE END OF ITS PRIMARY USE. NOT RECOMMENDED FOR USE IN AREAS WHERE VEGETABLES ARE GROWN.

TIP: ORDER 5-10% MORE LUMBER DEPENDING ON OVERALL QUANTITIES TO ALLOW FOR SOME VARIABLE LUMBER QUALITY AND INCORRECT CUTS.

NOTE: EXTERIOR WOOD TREATMENT CAN PROLONG THE LIFE OF YOUR PARK ELEMENTS. HOWEVER, IT IS NOT NECESSARY TO BRING MANY YEARS OF ENJOYMENT. ANY TREATMENTS MUST BE APPLIED AND MAINTAINED PROPERLY WHICH IS A LABOR INTENSIVE PROCESS. ONE SHOULD DECIDE IF THIS IS PRACTICAL OR NOT FOR THEIR PARK. THE PARK ELEMENTS ARE LOW COST, DURABLE, AND EASY TO BUILD BY DESIGN AND IT MAY BE MORE PRACTICAL TO SIMPLY HAVE THE NEXT GENERATION REBUILD THE ELEMENTS WHEN DEEMED NECESSARY. BELOW ARE RESOURCES FOR CONSIDERATION IF INTERESTED IN EXTERIOR WOOD TREATMENTS.

<https://www.rockler.com/learn/choosing-the-best-finish-for-outdoor-projects>

<https://www.realmilkpaint.com/blog/tips/best-finish-outdoor-wood/>

<https://vermontnaturalcoatings.com/product-category/exterior/>

4' GABION BENCH \

LUMBER:

(4.5) 2" x 4" x 8'

1 STEP

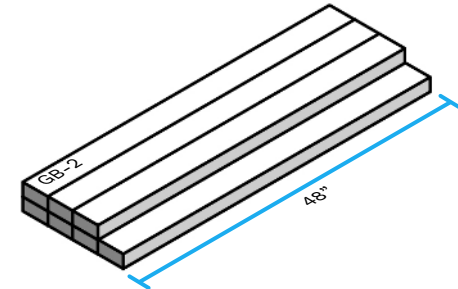
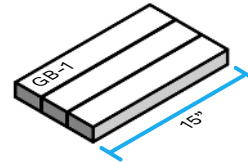
**1. MARK, LABEL AND CUT LUMBER
IN ORDER OF GB-1 TO GB-2 TO THE
FOLLOWING LENGTHS:**

NOTE: CHECK EACH PIECE OF LUMBER
IMPERFECTIONS SUCH AS CRACKS, WARPING, OR
UNEVEN MILLING. DISCARD THESE PORTIONS OF
LUMBER.

GABION BENCH ASSEMBLY

GB-1: (3) 2 x 4 x 15"
GABION BENCH FRAME

GB-2: (7) 2 x 4 x 48"
GABION BENCH TOP +
FRAME

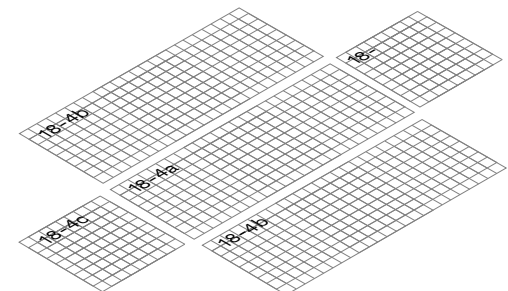
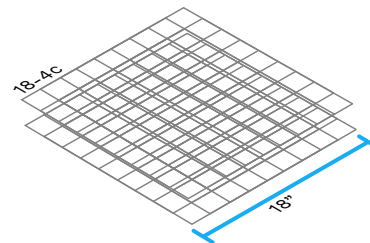
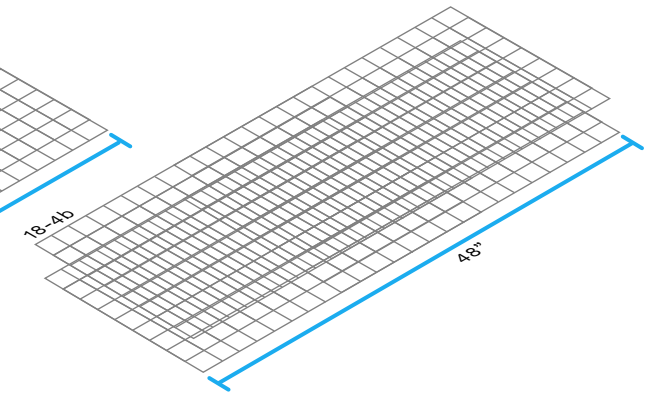
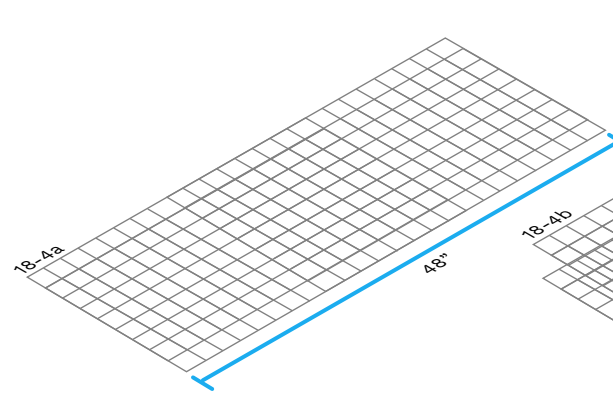


**2. SORT YOUR 2x2 WIRE MESH
INTO EQUAL SIZED PILES AND LABEL:**

18-4a (1) 48" x 18"
GABION BOTTOM

18-4b: (2) 48" x 18"
GABION SIDE LONG

18-4c: (2) 18" x 18"
GABION SIDE SHORT



BENCH MESH ADJACENCIES

TIPS:

- LABEL EACH PIECE OF LUMBER AS YOU ARE MEASURING/CUTTING
- HAVE A PILE OF LUMBER FOR GB-1 TO GB-2
- LIGHTLY SANDING CUT EDGES TO REMOVE SPLINTERS WILL ALSO HELP WITH ASSEMBLY.
- IF BUILDING MULTIPLE GABIONS, CREATING A STOP FOR YOUR MITER SAW ENSURES ALL PIECES ARE CUT TO THE SAME LENGTH

2 STEP

1. PLACE (2) GB-2 PIECES AND (2) GB-1 ON FLAT SURFACE. ALIGNING THE EDGES AS SHOWN IN THE IMAGE.

2. ENSURE THE VERTICALS ON THE GABION WIRE MESH ARE TOUCHING THE WOOD, AND THE CROSS PIECES (SMOOTH EDGES) THAT RUN THE SAME WAY AS THE BOARD ARE ON THE OUTSIDE (LIGHT BLUE), TO AVOID SHARP EDGES ON THE OUTSIDE OF THE GABION, AS SHOWN IN BLOW UP DIAGRAM.

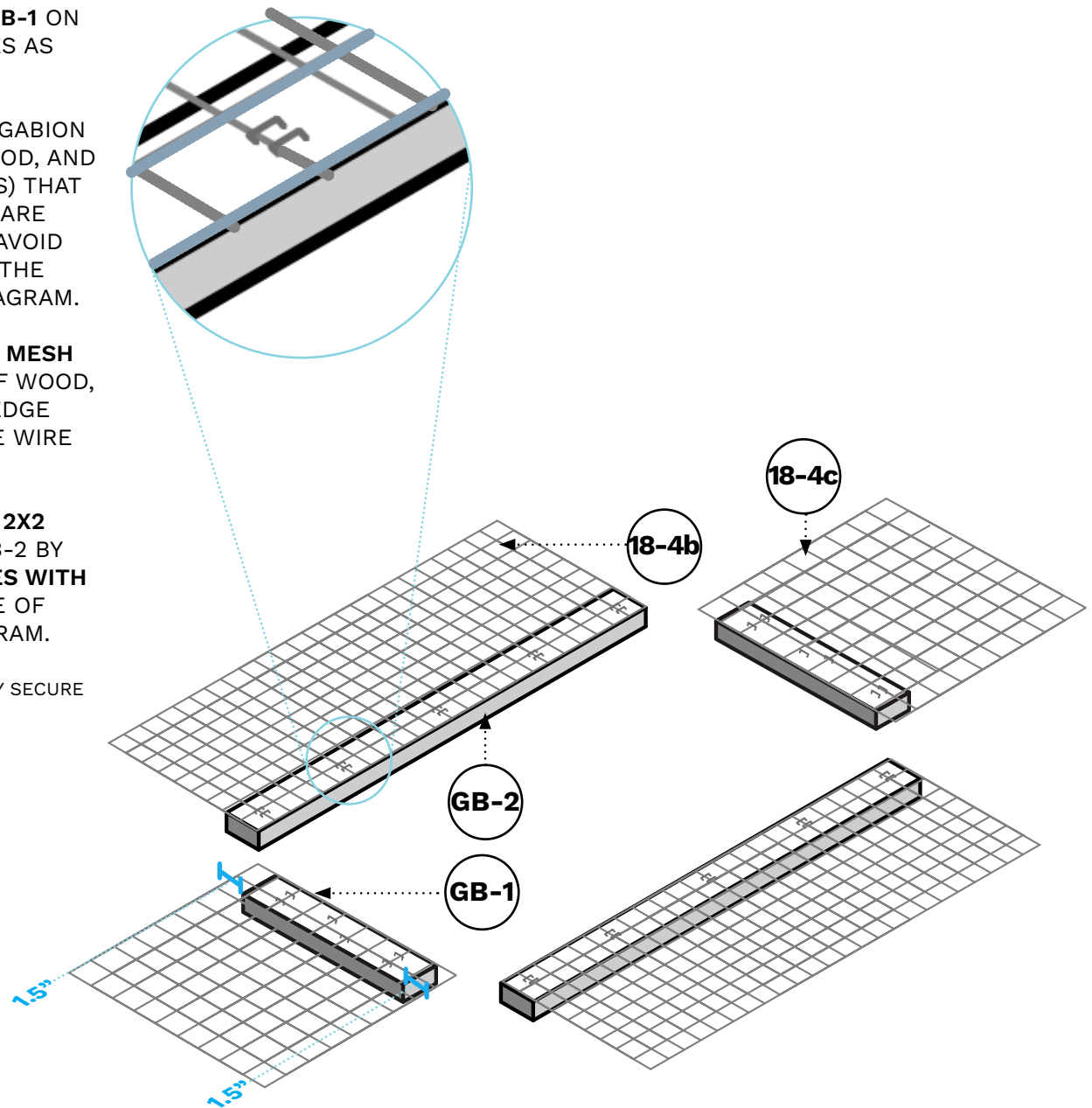
3. ENSURE THE (2) 18-4c 2X2 WIRE MESH IS CENTERED ON THE GB-1 PIECE OF WOOD, THERE SHOULD BE 1.5" FROM THE EDGE OF THE WOOD TO THE EDGE OF THE WIRE MESH.

4. SECURE (2) 18-4c AND (2) 18-4b 2X2 WIRE MESH ONTO THE GB-1 AND GB-2 BY USING (32) 1.5" GALVANIZED STAPLES WITH HAMMER. USE 2 STAPLES PER PIECE OF WIRE, AS SHOWN IN BLOW UP DIAGRAM.

NOTE: IF CONNECTING MANY GABIONS, ONLY SECURE THE METAL MESH TO (1) GB-1



1.5" CABLE STAPLE



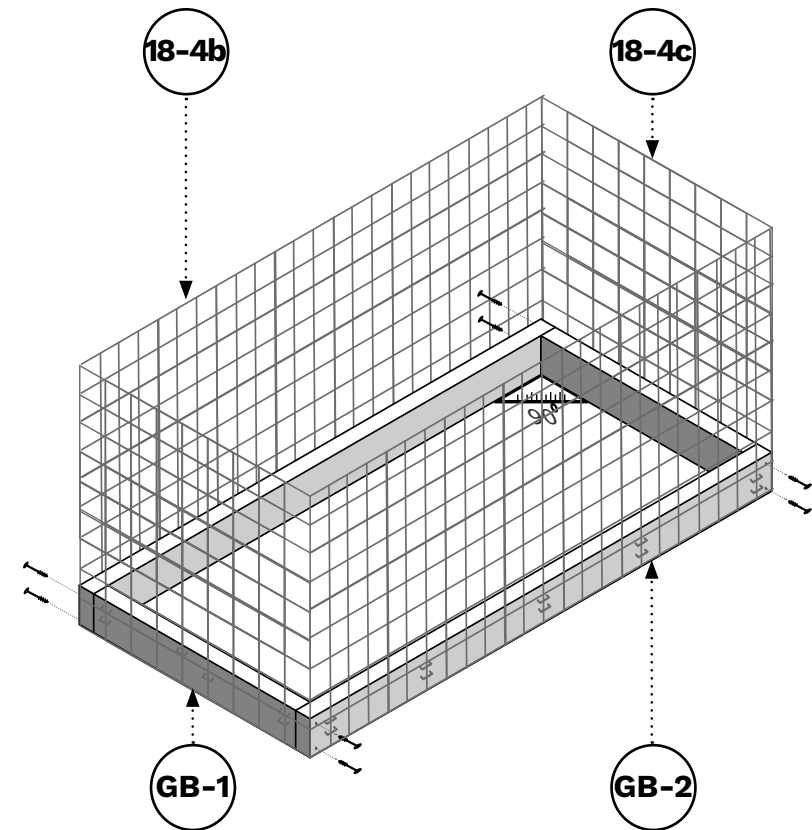
3 STEP

1. PLACE (2) **GB-2** PIECES ON FLAT SURFACE.
2. PLACE (2) **GB-1** IN BETWEEN THE **GB-2** PIECES ON EITHER END, ALIGNING THE EDGES AS SHOWN IN THE IMAGE.

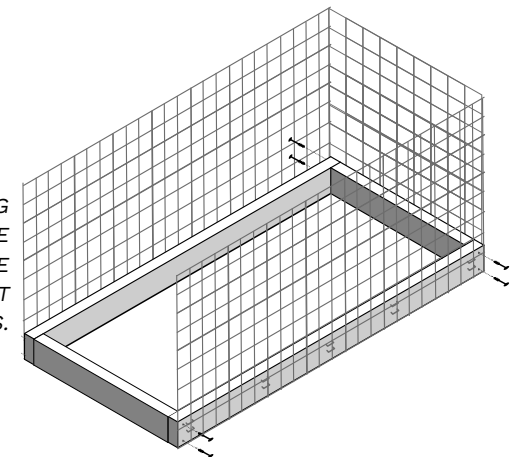
NOTE: UTILIZE A CARPENTERS SQUARE TO ENSURE A RIGHT ANGLE IS FORMED IN EACH CORNER. IF THE RIGHT ANGEL IS FAR OFF, CHECK THAT YOUR MEASUREMENTS OF EACH BOARD ARE CORRECT

3. USE THE **PRE-DRILLING GUIDE**  AT THE END OF THE PACKET TO PRE-DRILL YOUR SCREW HOLES

4. SECURE **GB-2** LUMBER INTO THE **GB-1** PIECES BY USING AN IMPACT DRIVER WITH (8) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS.

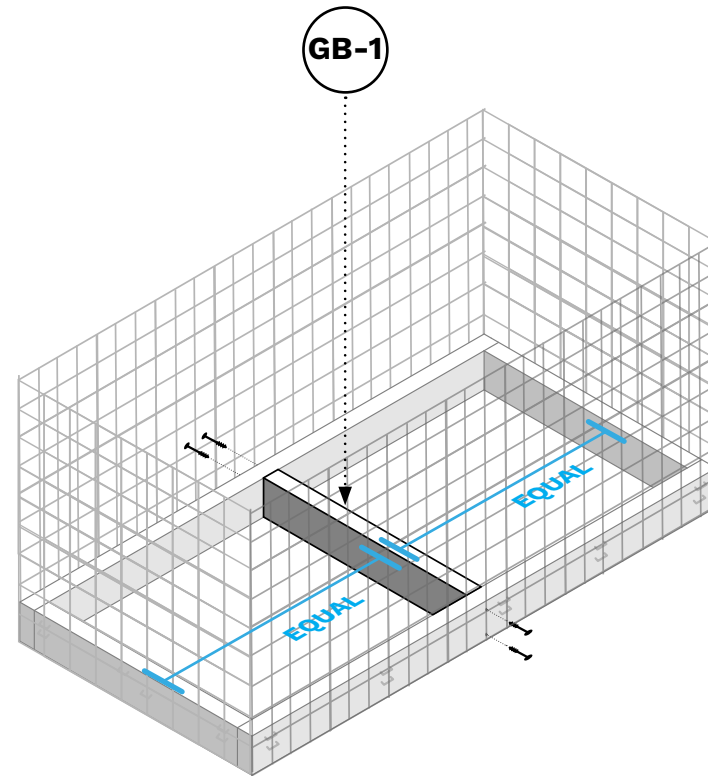


NOTE: IF YOU ARE CONNECTING MANY GABIONS TOGETHER, LEAVE ONE END OPEN. LOOK AT THE BUILD CHAPTER FOR LAYOUT INSTRUCTIONS.



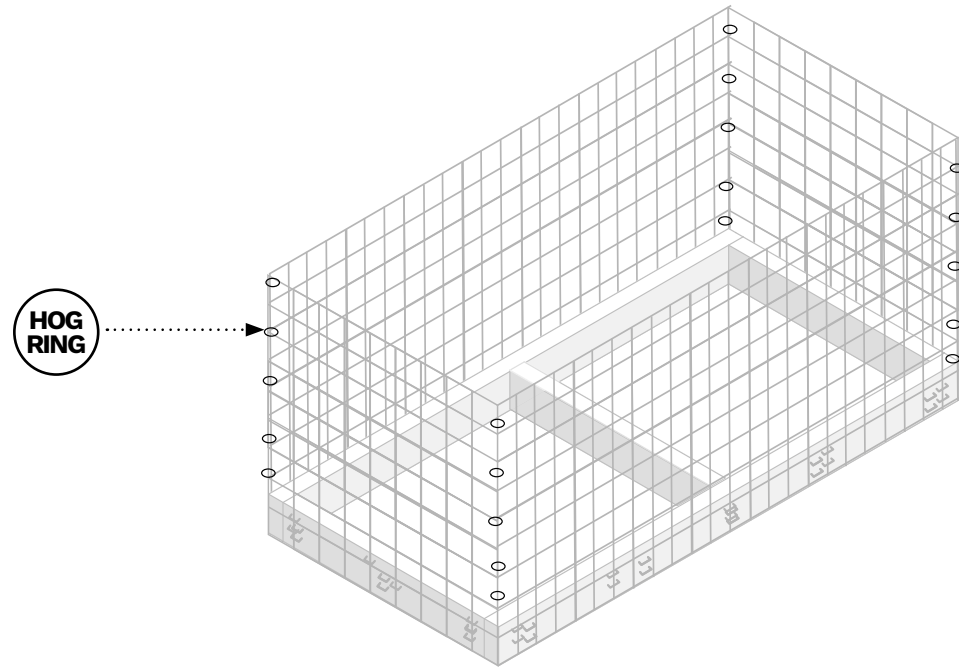
4 STEP

1. PLACE (1) **GB-1** PIECE IN THE MIDDLE OF THE FRAME CREATED IN STEP TWO.
2. SECURE LUMBER USING AN IMPACT DRIVER WITH (4) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS.



5 STEP

1. SECURE 2X2 WIRE SHEET USING (2) 18-4c AND (2) 18-4b PIECES TOGETHER BY USING HOG RINGS + HOG RING PLIERS TO SECURE BASKET TO ITSELF, ONE HOG RING EVERY OTHER OPENING. ENSURE HOG RINGS ARE AT ALL CORNERS AND EDGES



6 STEP

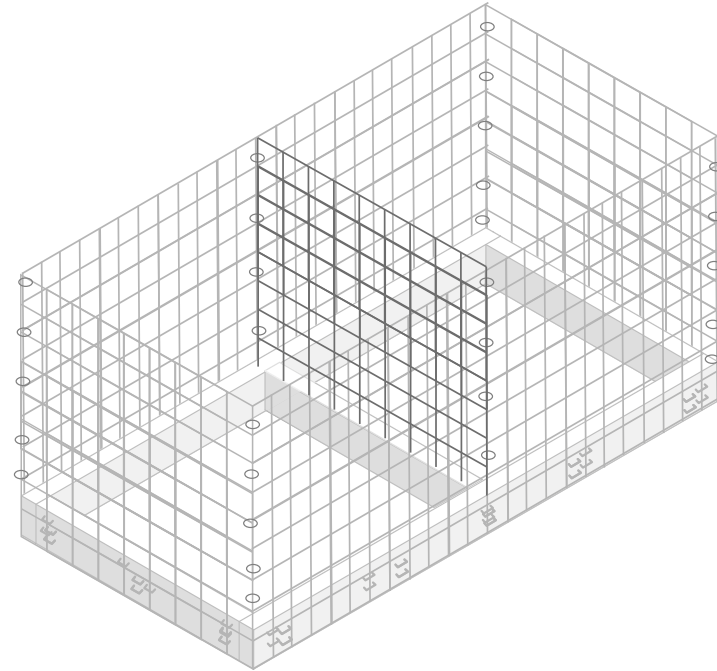
1. USE BRACING MATERIAL

2. IN THE CENTER OF THE GABION BASKET, BELOW THE MIDDLE GB-1 PIECE, USE ONE OF THE FOLLOWING OPTIONS:

OPT A: TAKE EXCESS END PIECES OF 2X2 WIRE METAL MESH + CUT IN HALF WITH BOLT CUTTERS

OPT B: USE ANY EXCESS 2X2 METAL MESH THAT SPANS THE ENTIRE WIDTH OF THE BASKET, SO LONG AS IT HAS A MINIMUM OF 2 BOXES IN HEIGHT (4"+ INCH MINIMUM)

2. SECURE **MIDDLE SUPPORT PIECE** BY USING **HOG RINGS**

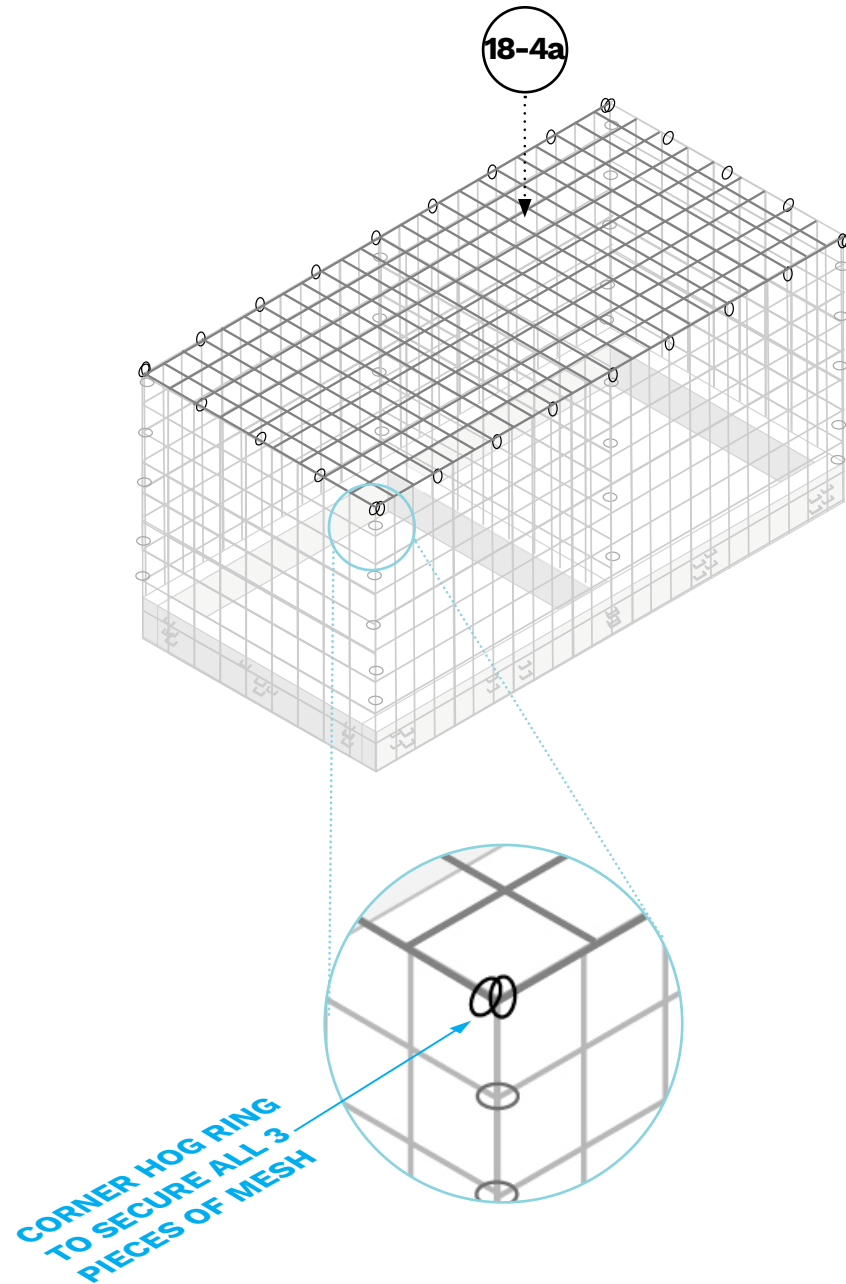


7 STEP

1. PLACE (1) 18-4a 2X2 WIRE MESH TOP PIECE ON TOP OF THE GABION BASKET SECURED IN STEP FIVE.

2. SECURE 2X2 WIRE MESH TOP PIECE BY USING THE HOG RINGS ON THE FOUR SIDES AND THE CENTER BRACE. SECURE 1 HOG RING EVERY OTHER MESH OPENING AND (2) AT EACH CORNER

NOTE: GABION BASKET IS NOW READY TO TRANSPORT TO SITE



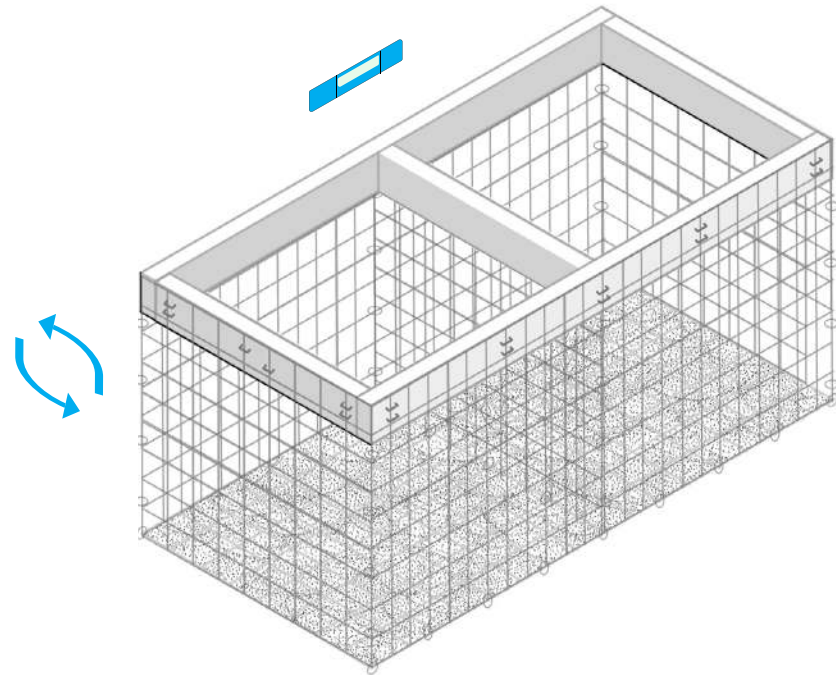
8 STEP

1. ROTATE GABION BASKET 180°.
2. PLACE ON-SITE WITH GRAVEL UNDERLAY. SEE FIG 1
3. ENSURE GABION BASKET IS STRAIGHT USING A LEVEL.

NOTE: IF NEXT TO OTHER GABIONS, USE STRING AND LINE LEVELS TO ENSURE GABIONS ARE STRAIGHT



FIG 1
LEVEL GABION BASKET
WITH GRAVEL



9 STEP

1. PLACE GABION IN ITS LOCATION PER YOUR PARKS DESIGN. APPLY GRAVEL IN BOTTOM TO FINE TUNE ADJUSTMENTS, SEE FIG 1 IN STEP 8.

NOTE: REFERENCE PHASE FOUR IN YOUR CREATE WORKBOOK.

3. STARTING WITH THE LARGEST, FLATTEST PIECES OF GABION FILL YOU HAVE, BEGIN FILLING GABIONS, STARTING AT THE EDGES. CONTINUE UNTIL YOU HAVE BUILT UP THE SIDES, SEE FIG. 2

4. FILL THE FRAME WITH LEFTOVER **GABION FILL** UNTIL IT IS LEVEL WITH THE TOP OF THE WOOD EDGE.

NOTE: CAREFULLY PLACE FILL INSIDE BASKET TO AVOID WARPING AND ENSURING ALL GAPS ARE FILLED FOR STABILITY

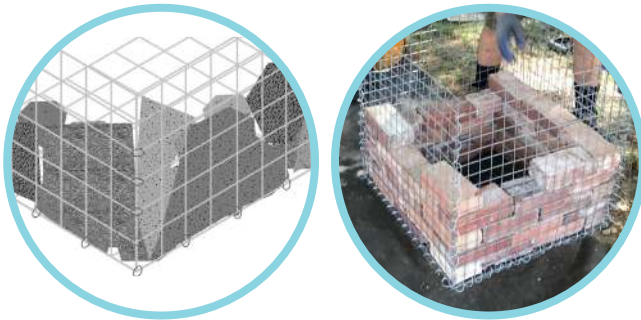
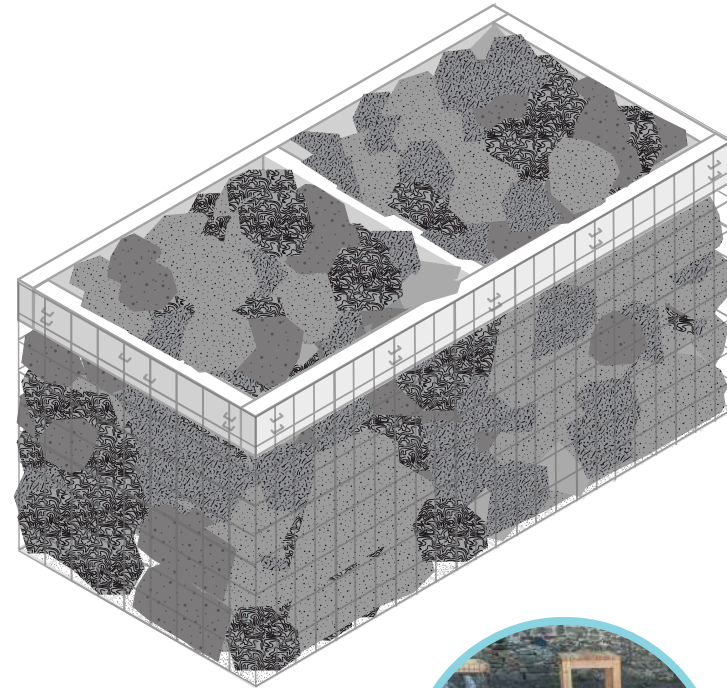


FIG 2

GABION BASKET FILLING ALONG EDGES FIRST



10 STEP

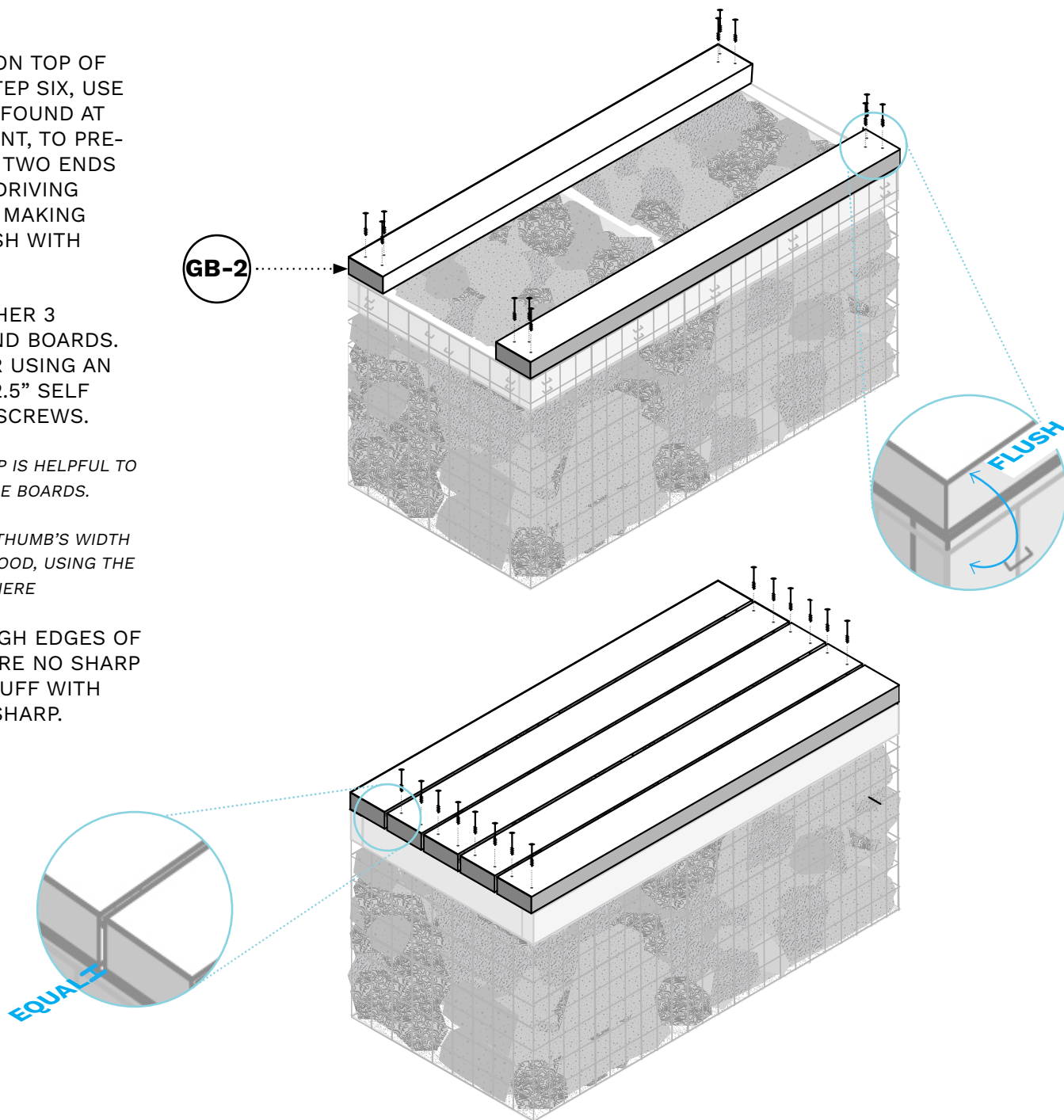
1. PLACE (2) GB-2 PIECES ON TOP OF THE FRAME CREATED IN STEP SIX, USE THE PRE-DRILLING GUIDE, FOUND AT THE END OF THIS DOCUMENT, TO PRE-DRILL THE WOOD. SECURE TWO ENDS FIRST WITH (12) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS, MAKING SURE TO ALIGN EDGE FLUSH WITH WOOD BELOW.

2. EQUALLY SPACE THE OTHER 3 BOARDS BETWEEN TWO END BOARDS. SECURE THE GB-2 LUMBER USING AN IMPACT DRIVER WITH (12) 2.5" SELF DRIVING EXTERIOR WOOD SCREWS.

NOTE: PREDRILLING AT THIS STEP IS HELPFUL TO ENSURE SCREWS DON'T SPLIT THE BOARDS.

TIP: ALL SCREWS SHOULD BE A THUMB'S WIDTH AWAY FROM THE EDGE OF THE WOOD, USING THE PRE-DRILLING GUIDE CAN HELP HERE

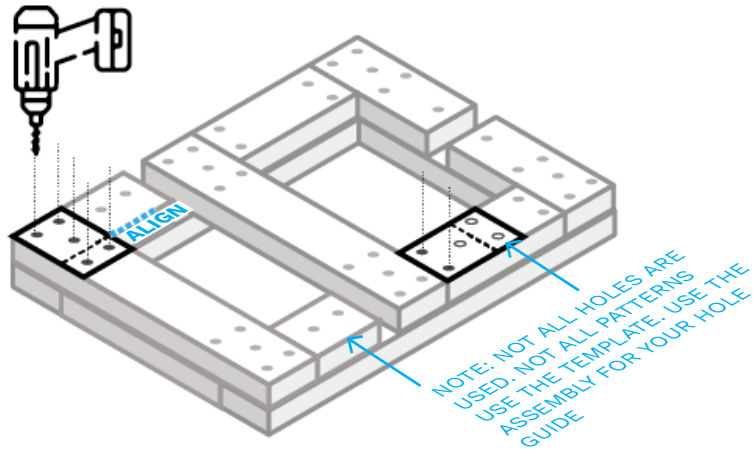
3. LIGHTLY SAND THE ROUGH EDGES OF THE GABION BENCH. ENSURE NO SHARP EDGES OF GABION WIRE, BUFF WITH GRINDER IF ANY ARE TOO SHARP.



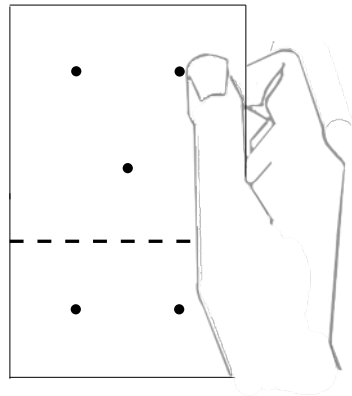


1. CUT OUT PRE-DRILLING HOLE GUIDE ALONG DOTTED LINES

2. WHEN DRILLING, ENSURE THAT YOU PROPERLY **ALIGN THE EDGE OF THE TEMPLATE WITH OUTSIDE EDGE OF THE LUMBER** BEFORE PRE-DRILLING



3. ENSURE ALL HOLES ARE ABOUT A **THUMB IN WIDTH FROM THE EDGE (~1")** EVEN HOLES THAT DON'T USE THIS TEMPLATE



PRE-DRILLING HOLE GUIDE

